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Review

The Current Generation of Tabular Foundation Models: A Critical Review

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Abstract

Tabular foundation models (TFMs) have moved tabular machine learning from per-dataset training towards amortised in-context inference, fitting a small-to-medium table in a single forward pass without a training run. The 2024–2026 release train, the TabPFN and TabICL lines and challengers such as Mitra, LimiX and Orion, has produced a generation whose architectures, capabilities and limits are documented mainly in preprints, while existing surveys treat these models as a subsection of tabular deep learning or of language-model table understanding. This review is, to our knowledge, the first organised around the current generation. From a corpus of 961 screened records and 98 retained studies, it taxonomises the architectures by pretraining regime, maps the capability space across five axes, isolates the language-model-on-tabular strand for prediction, feature engineering and generation, and summarises openness and deployment. A dedicated critical synthesis then reads the reported capabilities against independent evidence: on the studies reviewed here, tree-based and deep models retain the lead across 142 curated datasets that go beyond the standard independent and identically distributed setting; on 112 datasets, the models attain the highest accuracy but weaker conditional coverage than gradient-boosted trees; and robustness under feature shift, fairness and generation quality remain open. Amortised in-context prediction is thus a working paradigm whose independent evidence has yet to match its benchmark claims.

Keywords: tabular foundation models; in-context learning; prior-data fitted networks; TabPFN; tabular deep learning; large language models; synthetic data generation; benchmark evaluation; calibration; fairness

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1. Introduction

Tabular data remains the dominant modality of applied machine learning: most predictive workloads in industry, medicine, finance and public administration arrive as tables of mixed numerical and categorical features. For a decade the accepted answer was per-dataset training, with gradient-boosted decision trees as the reference method and a succession of specialised deep architectures competing against them with mixed success [1–14]. Every new dataset therefore required its own training run, hyperparameter search and evaluation protocol, and the cost of this cycle set the practical limits of tabular machine learning in small-data and rapid-turnaround settings.

Tabular foundation models (TFMs) change the unit of work. A transformer pretrained on a large distribution of synthetic or real tasks performs supervised learning as amortised inference in context: the labelled training set and the unlabelled queries are consumed together, and predictions are produced in a single forward pass, without task-specific weight updates [15,16]. TabPFN demonstrated the recipe for small classification tasks [15], and the publication of its second version in *Nature* moved the paradigm from a research curiosity towards a credible default for small-data prediction [17].

What followed was a rapid release train. Between 2024 and 2026 the TabPFN line advanced through versions 2.5 and 3 [18,19], the TabICL line scaled in-context learning to substantially larger tables [20,21], and challengers appeared with distinct design choices: Mitra with mixed synthetic priors [22], LimiX [23], Orion with multi-scale sparse attention [24], Google's TabFM [25], ConTextTab with semantics-aware context encoding [26], and real-data pretraining lines such as TabDPT and Real-TabPFN [27,28]. A parallel strand adapts large language models to tabular prediction, feature engineering and data generation [29–31]. The evidence about this generation is scattered across preprints, technical reports and vendor evaluations, and much of it postdates the surveys on which the field currently relies.

Existing surveys do not treat this generation as a subject in its own right. The representation-learning survey of Jiang et al. [32] and the tree-to-deep-learning surveys of Somvanshi et al. and Borisov et al. [1,33] cover the whole tabular deep-learning landscape and reach the foundation models only as the final stage of that progression or in passing, and the survey of Fang et al. [34] centres on language-model pipelines over tables and predates the current release train. None of them is organised around what the current models can do, how their architectures differ, and where independent evidence departs from the reported results. A dedicated review of the TFM generation is missing.

This review closes that gap with five contributions: (i) a framing of the TFM paradigm, its mechanism and its boundaries (Section 3); (ii) an architecture taxonomy of the 2024–2026 generation, organised by pretraining regime and adaptation mechanism (Section 4); (iii) a capability map covering context construction and scaling, mechanics and inductive bias, output modalities, interpretability and adaptation (Section 5); (iv) a positioning of the LLM-on-tabular strand for prediction, feature engineering and generation relative to dedicated TFMs (Section 6); and (v) a critical synthesis of the independent evidence on generalisation, calibration, robustness under feature shift and fairness, closed by an agenda of open problems (Section 8). The thread that runs through these contributions is a single reading of the generation: the current models have established amortised in-context prediction as a working paradigm, while the independent evidence on generalisation, calibration, robustness and fairness has yet to match its benchmark claims.

Table 1 sets this review against the three surveys that currently cover the ground, and Table 2 records the same comparison as a coverage matrix. The three treat the tabular landscape broadly and give the foundation-model generation a subsection, or, in the case of Fang et al., centre on language-model pipelines and predate the current release train. None is organised around the capabilities, the architectural differences and the independent evidence of the 2024–2026 generation, which is the ground the present review occupies; Table 2 shows that this review is the only one of the four to fill every column.

The scope is tabular data only, covering prediction and generation; table question answering, NL2SQL, table visualisation and document extraction concern a different modality of work over tables and are excluded throughout. The present article is organised by capabilities and architectures; a study-level audit of evaluation rigour in this literature is a separate undertaking and is outside its scope. Section 2 documents how the literature

base was assembled, Section 7 summarises applications and deployment considerations including openness and licensing, and Section 9 concludes.

Table 1. This review set against the surveys that currently cover the tabular foundation model literature. The final column states how the present review departs from each prior survey; its own row states the review's contribution.

Prior Survey	Scope	Organising Principle	Limitation for the 2024–2026 Generation	Departure of This Review
Jiang et al. [32]	Tabular deep-learning representation learning	By representation-learning method	Foundation models enter as one late stage of the method progression, reaching TabPFNv2 and TabICL, without a dedicated account of the current generation	Treats the current generation as the subject, organised by capability and architecture
Somvanshi et al. [1]	Tree-based and deep tabular models	By model family, from gradient-boosted trees to deep networks	No architecture taxonomy of the foundation-model generation and no independent-evidence synthesis	Adds an architecture taxonomy and a synthesis of the independent evidence
Fang et al. [34]	Language models over tables, for prediction, generation and understanding	By task, technique and metric	Centres on language-model pipelines, not dedicated foundation models, and predates the current release train	Isolates the language-model-on-tabular strand (Section 6) and relates it to dedicated foundation models
This review	The dedicated foundation-model generation, 2021–2026, for prediction and generation	By capability and architecture	Narrative rather than systematic, so no completeness claim of the PRISMA kind	First dedicated account of the current generation with a synthesis of the independent evidence

Table 2. Coverage matrix of the same four works. ● = covered as a primary subject; ○ = covered partially or as a subsection; — = not covered. Abbreviations: Gen. = the 2024–2026 model generation; Tax. = architecture taxonomy of the generation; Cap. = capability map of the generation; LLM = language-model-on-tabular strand; Evid. = synthesis of independent evidence; Open. = openness and licensing. Coverage windows are approximate.

Work	Gen.	Tax.	Cap.	LLM	Evid.	Open.	Coverage
Jiang et al. [32]	○	—	—	○	—	—	≤2025
Somvanshi et al. [1]	○	—	—	○	—	—	≤2026
Fang et al. [34]	—	—	—	●	—	—	≤2024
This review	●	●	●	●	●	●	2021–2026

2. Scope and Review Methodology

This is a narrative critical review rather than a systematic one, and we therefore make no completeness claim of the PRISMA kind; we nevertheless document how the literature base was assembled so that the coverage can be audited. A Scopus title–abstract–keyword query restricted to publication years after 2022 identified 851 records. Because the defining releases of the current generation appear as preprints, technical reports and model-repository documentation before, or instead of, indexed publications, the indexed set was extended by seed papers known to the authors and by a targeted search of arXiv, OpenReview and model repositories; after de-duplication the screening corpus held 961 records.

Records were screened in two passes against the scope stated in Section 1. Work on table question answering, NL2SQL, table visualisation and information extraction from documents was excluded explicitly (91 records), as were per-dataset architectures without a pretraining component and application studies whose subject is the application domain and not the model family. The retained body comprises 98 studies, assigned to the thematic sections of this review; the remaining records provide context. Screening and thematic

assignment were tool-assisted and verified by the authors, and the corpus with per-record decisions, the query string and the model registry underlying the model and openness tables of Sections 4 and 7 are deposited alongside the article (Section 9). Large language models were used as assistive tools for literature screening and draft editing under author verification, and this use is declared in the back matter.

Although the review is narrative, several elements of systematic practice were adopted so that the synthesis can be audited and repeated. The database query, its date and its year restriction are stated above and reproduced in Appendix A; inclusion and exclusion criteria were fixed before screening and applied in two passes; the exclusions are reported with counts by reason; and the per-record decisions are deposited so that a reader can recompute the retained set. Three elements of a systematic review were deliberately not adopted, and their absence bounds what the corpus can support: screening was performed by one author and verified by a second on the disputed records instead of by two independent screeners with an agreement statistic; no formal risk-of-bias instrument was applied to individual studies, since no instrument in current use fits benchmark-driven machine-learning reports; and no registered protocol or PRISMA flow diagram accompanies the search. The principal threat this leaves is selection bias in two directions. Coverage favours the visible model lines because the targeted extension that recovers preprint-stage work was seeded by papers known to the authors, so a well-publicised release is more likely to be captured than an equally relevant obscure one. Coverage also favours reported success because independent evaluations that find no difference are less likely to be written and indexed than evaluations that find one. Both biases work in the same direction, towards a more favourable picture of the paradigm than the underlying literature may warrant, which is worth holding in mind against the critical reading in Section 8.

The maturity of the evidence base deserves a statement of its own because it qualifies every claim synthesised below. The field is young enough that a large share of its primary record has not been through peer review: of the 152 works cited here, 61, or two in five, are preprints or technical reports, and that share is concentrated in the most recent material, with 38 of the 61 dated 2025 or 2026. The model registry shows the same pattern, with the model releases surveyed in Section 4 split evenly between archival publications and preprints or vendor technical reports. Three consequences follow. Claims resting on vendor technical reports have not had an external check of their experimental protocol, so we treat them throughout as reported and not as established, and mark them as such in the claim-versus-evidence synthesis of Section 8. Preprint results can change between versions, which is why every model citation is pinned to a versioned source with its access date. And the independent evaluations that anchor Section 8 are themselves recent, several of them preprints, so the critical reading is provisional in the same way as the claims it examines. Readers should therefore treat this review as a synthesis of the evidence available in mid-2026 and not as a settled account of the field.

Two further consequences of the assembly procedure bound the interpretation of what follows. Coverage of work released after the search date reflects the targeted extension and not the indexed query, so recency is better for the major model lines than for the long tail; and because a large share of the primary evidence is preprint-stage, we cite preprint versions with their version numbers and dates, and treat vendor-reported results as claims to be read against the independent evaluations synthesised in Section 8.

3. The Tabular Foundation Model Paradigm

We use the term tabular foundation model for a model that satisfies three conditions: it is pretrained across a distribution of tasks rather than on a single dataset; at deployment it consumes a labelled training set and unlabelled queries together in its context and produces

predictions for the queries in a forward pass; and it requires no task-specific weight updates to do so, although it may accept them as an option. The prior-data fitted network (PFN) formulation makes the statistical reading of this mechanism explicit: pretraining on tasks drawn from a prior teaches the network to approximate the posterior predictive distribution of that prior, so in-context prediction is amortised Bayesian inference [15,16]. On this reading the pretraining prior plays the role that model class and regularisation play in classical learning, and much of the architectural variation surveyed in Section 4 is variation in how that prior is constructed and encoded.

Figure 1 draws the contrast that defines the generation: the classical route fits and tunes a model on each dataset in turn, while the amortised route performs that fitting once during pretraining and spends inference on a single forward pass over the labelled context and the query. The rest of the review follows the consequences of moving the work from per-dataset training to a shared prior, for what the models can do, what they cost to run, and where independent evidence departs from the reported strengths.

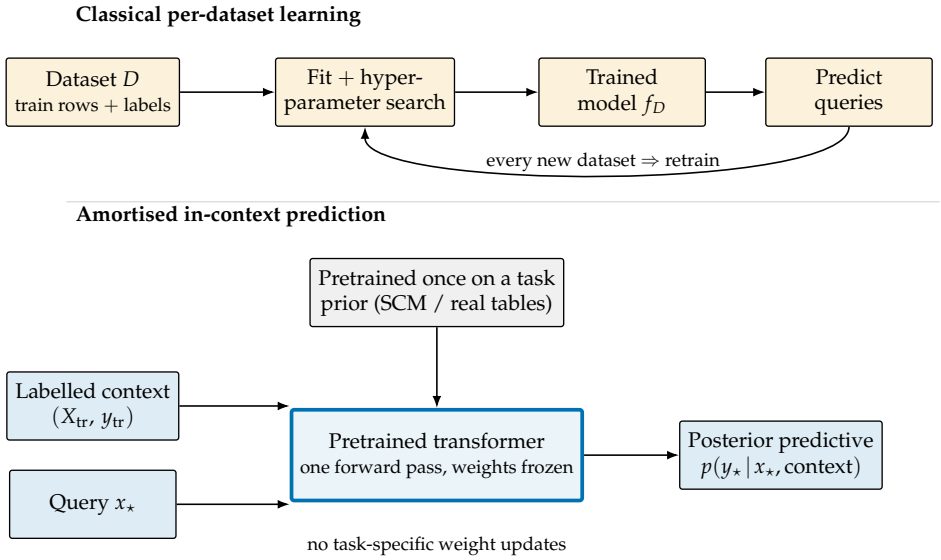


Figure 1. Amortised in-context prediction contrasted with classical per-dataset learning. In the classical route a model is fit and tuned on each dataset in turn; in the amortised route a transformer, pretrained once on a distribution of tasks, consumes the labelled context and the query together in a single forward pass and emits a posterior predictive without task-specific weight updates.

The lineage of the mechanism is short. Non-parametric transformers introduced attention between datapoints, treating the dataset rather than the row as the object the network reasons over [35]. That a transformer can be trained to carry out a learning algorithm in its forward pass, beyond storing a fixed input–output mapping, was established for controlled function classes and read as the in-context implementation of a learning rule, close in behaviour to a step of gradient descent on the context [36–38]; the tabular models below inherit that reading and specialise it to tables. TabPFN combined this architectural core with pretraining on synthetic tasks drawn from structural causal models, and showed that the result is a practical classifier for small tables that needs no tuning [15]. The second version extended the recipe to regression, larger tables and missing data, and its publication in *Nature* gave the paradigm a visibility that the preprint literature had not [17]; the release train described in Section 4 followed within eighteen months.

Three enablers converged to make the generation possible. First, prior design matured: structural causal priors were joined by mixed and structured synthetic priors, and by pretraining on curated real tables [22,27,28]. Second, the engineering of context scaling

advanced, through architectural work on attention structure and through inference-time techniques examined in Section 5.1, lifting the original limits on rows and features [20,24]. Third, the surrounding infrastructure of benchmarks, leaderboards and tooling lowered the cost of demonstrating and adopting new models, a development whose consequences for independent evaluation are taken up in Section 8.6.

Terminology in the literature is unsettled, and we fix our usage here. Tabular in-context learning names the deployment mechanism; prior-data fitted networks name the synthetic-prior pretraining formulation; tabular foundation model names the model class, whatever the pretraining regime. The class excludes per-dataset deep architectures without cross-task pretraining, AutoML systems that search over conventional learners, and language-model pipelines whose subject is text over tables and not prediction or generation; the boundary with the last group is drawn in Section 6.

4. Architectures of Tabular Foundation Models

The models surveyed in this section share the deployment mechanism fixed in Section 3 and differ in construction. Two questions organise the differences: where the pretraining signal comes from, and how the pretrained model is adapted to a new task. The first separates synthetic priors (Section 4.1) from real-data and continued pretraining (Section 4.2) and from the earlier cross-table representation line (Section 4.3); the second separates pure in-context prediction from finetuning-based transfer and from mechanisms that amortise into something other than the context, such as hypernetworks (Section 4.4). Section 4.5 assembles the release timeline (Figure 2), the resulting taxonomy (Figure 3) and a reference summary of the current generation (Table 3).

4.1. Synthetic-Prior In-Context Learners

The synthetic-prior family is the direct continuation of the PFN recipe and remains the centre of gravity of the generation. Its development is anchored by two laboratories whose release lines have advanced in step. The TabPFN line moved from the original small-table classifier [15] to a second version whose alternating attention over columns and rows extended the recipe to regression, missing values and medium-sized tables [17], then to TabPFN-2.5, which raised the practical envelope to roughly 50,000 samples and 2000 features and reports parity with ensembles tuned for four hours [18], and to TabPFN-3, whose technical report describes scaling to one million rows and 200 features, a many-class decoder, cache-accelerated inference and an optional inference-time reasoning mode [19].

The TabICL line took the scaling question first. TabICL factorises the context into a column-embedding stage followed by row-wise in-context attention, which brings tables of several hundred thousand rows within reach of a single forward pass [20]; TabICLv2 improves the synthetic pretraining distribution and the architecture, extends the model to regression, and ships checkpoints under a permissive licence [21]. The two lines differ instructively in openness: the TabICL checkpoints and the v1 pretraining code are open, whereas the recent TabPFN releases place their weights under research-and-evaluation licences, a contrast Section 7.4 returns to.

The challengers are best read as experiments on the prior rather than on the architecture. Mitra keeps a plain twelve-layer in-context transformer and varies what it is pretrained on, mixing structural-causal-model tasks with tree-based generators; its reported ablations locate the gains on small tables in this mixture and not in architectural change [22]. LimiX couples hierarchical structural-causal priors with attention-guided retrieval over the context and extends the task set towards imputation and sample selection, with the stated ambition of a general structured-data model; its claim of leading results across ten benchmarks rests on the developers' own evaluation [23]. Google's TabFM, released as this review was

being finalised, joins the same class from the industrial side: it stacks TabPFN-style alternating row-and-column attention, a row-compression step, and a TabICL-style in-context transformer over the compressed rows, is pretrained entirely on hundreds of millions of synthetic structural-causal-model datasets, and ships classification (up to ten classes) and regression checkpoints under a non-commercial weights licence [25]. Its reported advantage over tuned gradient-boosted baselines on the 51 datasets of TabArena [39] is, at the time of writing, vendor-reported and awaits independent evaluation, and the model predates any derivative work, so it is recorded in the registry (Table 3) and the timeline (Figure 2) but does not yet appear in the capability map of Section 5.

Within this family the architectural core has converged while the priors diverge. Every model consumes the labelled context through some arrangement of attention across samples and features, and the differences in that arrangement are increasingly engineering choices in service of scale; what a model assumes about tables before it sees one, in the sense of Section 3, is decided by the prior. This concentration of design effort on the prior is the family's defining tension, and it carries an evidentiary consequence: prior construction is documented in technical reports of varying depth, and several headline results, including the TabPFN-3 leaderboard margin, are vendor-reported and await independent replication, a point Section 8 takes up.

4.2. Real-Data and Continued Pretraining

Synthetic priors purchase control at the price of realism. A generator supplies unlimited pretraining tasks, guarantees that no evaluation dataset leaks into pretraining, and lets the designer decide the causal structure of what the model learns. What a generator does not supply is the texture of real tables, in which column names, string categories and value distributions carry meaning that a structural-causal simulator does not produce. Real-data pretraining inverts the trade, and the models covered here occupy the resulting spectrum.

TabDPT pretrains on curated real tables and assembles the context by retrieval, showing that in-context prediction survives the move away from synthetic tasks and reporting scaling behaviour with model and data size [27]. Real-TabPFN keeps the synthetic foundation and adds a continued-pretraining stage on curated real-world corpora, a hybrid that preserves the coverage of the simulator while adapting the model to real distributions [28]. ConTextTab pushes furthest towards semantics: trained on a large real-world tabular corpus, it encodes cell contents and column headers with semantics-aware embeddings, so that a header string such as *temperature* is information rather than an arbitrary symbol [26].

Kim et al. frame the choice systematically as a question of knowledge pretraining, arguing that the semantic reading of real tables and the statistical reading of synthetic priors contribute complementary capabilities and examining how each can be embedded into a reusable model [40]. The move to real corpora also imports a question the synthetic family does not face: whether pretraining tables overlap with the benchmarks used for evaluation. That question belongs with the independent evidence and is taken up in Section 8.1.

4.3. Cross-Table Representation Pretraining

Before amortised in-context prediction became practical, transfer across tables was pursued through weights: pretrain a representation on many tables, then finetune on the target task [41–44]. The obstacle is schema heterogeneity since tables share neither columns nor column order, and the line is largely a sequence of answers to it. TransTab makes the transformer transferable across variable column sets [45]; XTab shares transformer blocks across datasets while keeping dataset-specific featurisers [46]; TabRet retokenises so that columns unseen in pretraining can be absorbed downstream [47]; UniTabE pretrains at the level of individual cells under a universal protocol [48]; and XTFormer casts the

problem as learning a meta-function space from which a mapping suited to the downstream table is extracted [49]. TabPTM stands slightly apart: its meta-representations support generalisation to unseen tables without task-specific training, which makes it the closest precursor within this line of the deployment contract the ICL generation adopted [50].

The idea from this line that survives into the current generation is language as the bridge across schemas. CARTE represents a table as a graph whose nodes carry textual embeddings of column names and entries, and pretrains a graph-attention network for context-aware transfer across heterogeneous tables [51]. TP-BERTa starts from a pretrained language model and adapts it to tabular prediction with relative-magnitude tokenisation of numerical values [52]. TabSTAR targets tables with free-text fields directly, treating textual columns as first-class input [53].

Read chronologically, the line peaks in 2022–2024 and is then absorbed. Its questions persist: unseen columns, header semantics and cross-table transfer reappear inside the ICL generation as questions of context encoding, and its text-awareness is carried forward by the real-data models of Section 4.2 and by TabSTAR. What changed is the deployment contract, since finetuning per task gave way to prediction in context; the models of this subsection are therefore evidence for the taxonomy of Section 4.5 rather than competitors of the models in Sections 4.1 and 4.2.

4.4. Alternative Mechanisms: Hypernetworks, Sparse Attention and Structured Encodings

A hypernetwork offers a different amortisation target. MotherNet is pretrained, like a PFN, across synthetic tasks, but its forward pass emits the weights of a small task-specific network rather than predictions; the child network then serves the task at negligible cost and without the parent in the loop [54]. The design decouples the expensive consumption of the context from serving, which matters wherever per-query transformer inference over the full training set is too slow, and it anticipates the deployment concerns of Section 7.3.

The remaining designs attack the cost and the encoding of attention itself. Orion-MSP replaces dense row attention with multi-scale sparse attention at window sizes 1, 4 and 16 and adds a Perceiver-style cross-component memory, extending in-context learning to tables with more than one hundred features [24]; its sibling Orion-BiX alternates standard, grouped, hierarchical and relational attention over the feature axis and aggregates the result through multiple class tokens [55]. Tab-PET derives positional encodings for tabular transformers from graph structure over the features [56]. TabSketchFM carries the toolkit to an adjacent task, pretraining sketch-based representations for data discovery over data lakes [57]. Additive tabular transformers constrain the architecture to a generalised-additive form; their interpretability consequences are examined in Section 5.4 [58].

4.5. A Taxonomy and the Release Train

The release chronology in Figure 2 makes the pace of the generation visible. The mechanism is older than the attention around it: attention between datapoints dates to 2021 [35] and the first TabPFN to 2022–2023 [15], while the cross-table transfer line of Section 4.3 ran in parallel through 2024. What changes from late 2024 onwards is the cadence: the two anchor laboratories begin shipping versions at intervals of months, and industrial entrants release competing models within a single year.

Two properties of the timeline matter for how the rest of this review reads the evidence. Most releases from 2025 onwards are documented in technical reports and preprints, with peer review pending or absent, so the primary description of the generation is largely vendor-authored; and the intervals between versions are shorter than a typical review cycle, so independent evaluations lag the models they evaluate by construction. Figure 3 abstracts the same population into the two axes fixed at the head of this section.

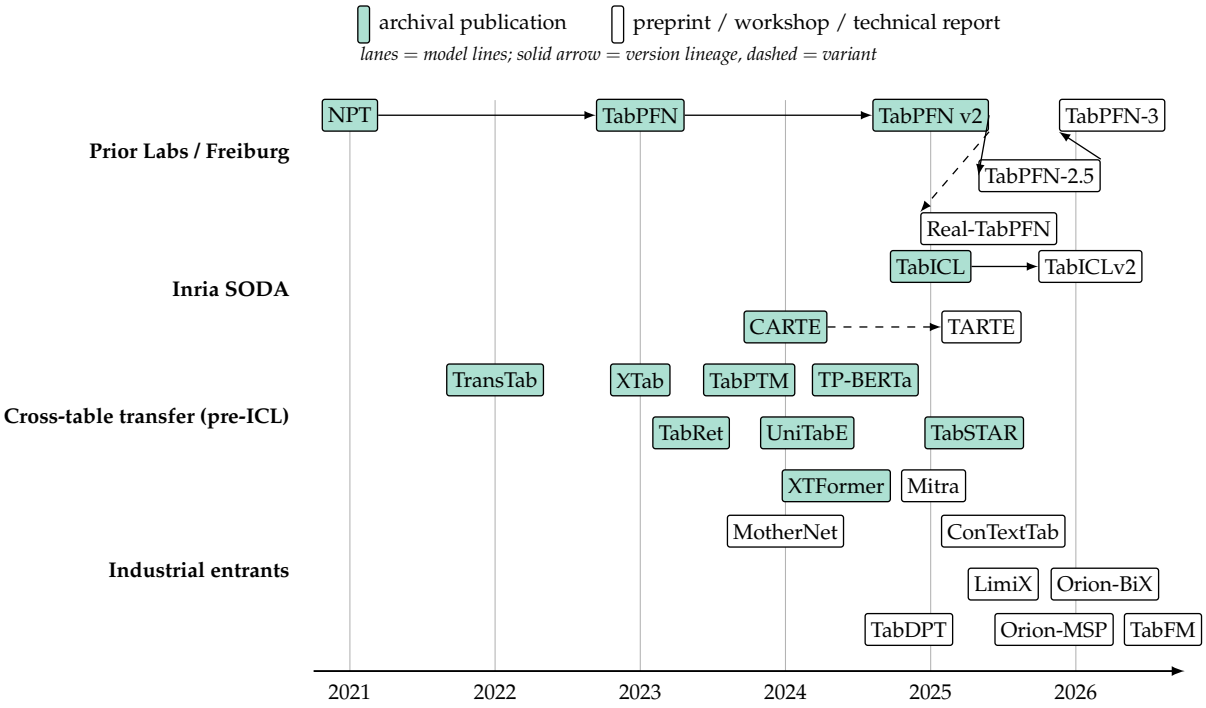


Figure 2. Release timeline of tabular foundation models, 2021–2026, grouped by laboratory or line. From late 2024 the anchor lines (Prior Labs; Inria SODA) ship versions at intervals of months and industrial entrants join within a single year; most post-2024 releases are documented as technical reports or preprints, without an archival publication. Dates are publication years of the versioned source cited in the text.

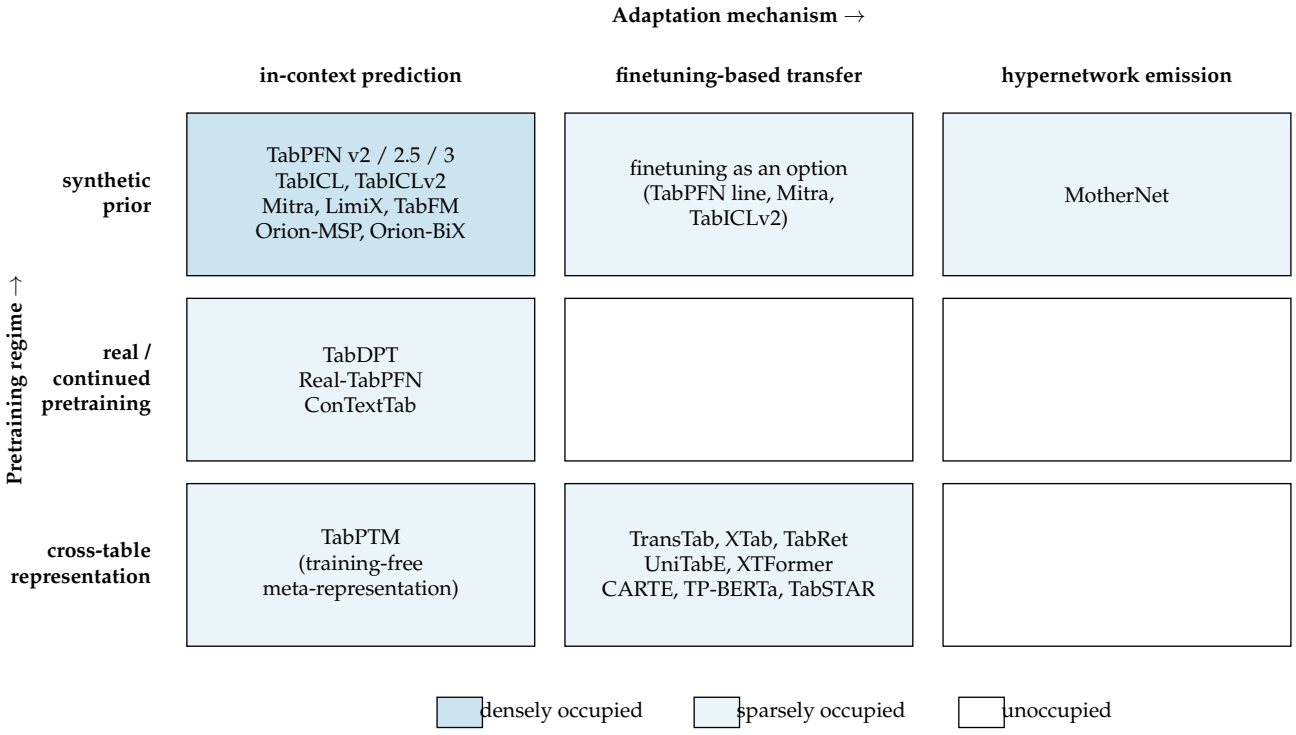


Figure 3. Architecture taxonomy of the generation: pretraining regime (rows) against adaptation mechanism (columns), with representative models in each occupied cell. The synthetic-prior in-context cell holds the anchor lines and most challengers; the real-data cells are filling rapidly; the hypernetwork cell remains a single line of work.

The taxonomy separates the pretraining regime (synthetic prior, real or continued pretraining, cross-table representation) from the adaptation mechanism (in-context prediction, finetuning-based transfer, hypernetwork emission), and its cells are unevenly occupied: the synthetic-prior in-context cell holds the anchor lines and most challengers, the real-data cells are filling rapidly, and the hypernetwork cell remains a single line of work. Table 3 records the current generation with the specifications that can be pinned to versioned sources; the pre-ICL transfer line of Section 4.3 appears in Figure 2 but predates the generation the table summarises.

Table 3. The current generation of tabular foundation models. Every specification is taken from the versioned source cited in the text or from the official model card or repository; repository and model-card facts were accessed on 13–14 July 2026. Abbreviations: cls = classification, reg = regression, imp = imputation; n/r = not reported in the cited versioned source; n/v = not verified at source at the time of writing. Reported scale is the developers' stated operating envelope, not an independently measured limit.

Model	Year	Developer	Pretraining	Tasks	Reported Scale	Params	Weights Licence
TabPFN v2 [17]	2025	Prior Labs	synthetic (SCM priors)	cls, reg	small–medium tables	n/r	Prior Labs licence (Apache-2.0 with attribution)
TabPFN-2.5 [18]	2025	Prior Labs	synthetic; Real -TabPFN-2.5 variant continues on real data	cls, reg	$\leq$50 k rows, $\leq$2 k features	n/r	tabpfn-2.5-license -v1.1 (research and internal evaluation; non-commercial)
TabPFN-3 [19]	2026	Prior Labs	synthetic	cls, reg	1 M rows, 200 features	n/r	tabpfn-3-license-v1.0 (non-commercial)
TabICL [20]	2025	Inria SODA	synthetic ($\leq$60 k-sample tasks)	cls	$\sim$500 k rows via memory-efficient inference	n/r	BSD-3-Clause
TabICLv2 [21]	2026	Inria SODA	synthetic (revised prior)	cls, reg	50 k$\times$100 in under 10 s on one GPU (reported)	n/r	BSD-3-Clause
Mitra [22]	2025	Amazon (AutoGluon)	synthetic mixed priors (SCM and tree generators)	cls, reg	strongest below 5 k rows, 100 features (reported)	72 M	Apache-2.0
LimiX [23]	2025	Stable AI & Tsinghua	synthetic hierarchical SCM priors	cls, reg, imp	n/r	16 M/2 M variants	code Apache-2.0; weights academic-open, commercial use by authorisation
TabFM [25]	2026	Google	synthetic (SCM priors, hundreds of millions of tables)	cls ($\leq$10 classes), reg	TabArena range 700–150 k rows (reported)	n/r	code Apache-2.0; weights TabFM Non-Commercial License v1.0
TabDPT [27]	2025	Layer6 AI	real tables, retrieval-assembled context	cls, reg	n/r	n/r	Apache-2.0
Real-TabPFN [28]	2025	Prior Labs	synthetic, continued pretraining on real corpora	cls, reg	n/r	n/r	n/v
ConTextTab [26]	2025	SAP	real corpus (T4 subset of TabLib)	cls, reg	n/r	n/r	Apache-2.0; checkpoints inherit research-only corpus terms
MotherNet [54]	2024	Microsoft	synthetic (PFN-style)	cls (child net)	n/r	n/r	research prototype (microsoft/ticl)
Orion-MSP [24]	2025	Lexsi Labs	synthetic	cls	beyond 100 features (reported)	n/r	MIT (announced)
Orion-BiX [55]	2026	Lexsi Labs	synthetic causal priors, episodic meta-training	cls	n/r	n/r	MIT (announced)

Two columns of Table 3 carry weight later in the review. The licence column splits the generation into permissively licensed models (TabICL and TabICLv2, Mitra, TabDPT, the Orion line) and models whose weights are restricted to research and evaluation (the recent TabPFN releases), tied to a restricted pretraining corpus (ConTextTab's checkpoints) or open for academic use only (LimiX); the consequences of this split for independent benchmarking and applied adoption are drawn out in Section 7.4. The sources column of the underlying registry is dominated by versioned technical reports, few of them archival, which is why every specification in the table is pinned to a versioned document with its access date, and why the capability claims mapped in Section 5 are read against the independent evidence of Section 8.

5. Capabilities and Mechanics

Sections 3 and 4 fixed what a tabular foundation model is and how the current generation is constructed. This section asks what the generation can do, through which mechanism, and at what cost. We read the capabilities along five axes: the construction and scaling of the in-context training set (Section 5.1), the mechanics and inductive bias of the forward pass (Section 5.2), the breadth of tasks and output modalities the mechanism has been carried to (Section 5.3), interpretability (Section 5.4), and adaptation beyond frozen in-context prediction (Section 5.5). These five axes are the measuring rod of the review. Section 8 takes each of them again and asks how far the reported capability survives independent evaluation, so a capability recorded here as available is a claim to be tested there, not a settled result. Reading the generation by capability, and not model by model, is what allows the same axes to carry the reported strengths of this section and the independent limits of Section 8 without restating the argument; where a capability is distributed unevenly across the generation, Figure 4 records which models occupy which axis.

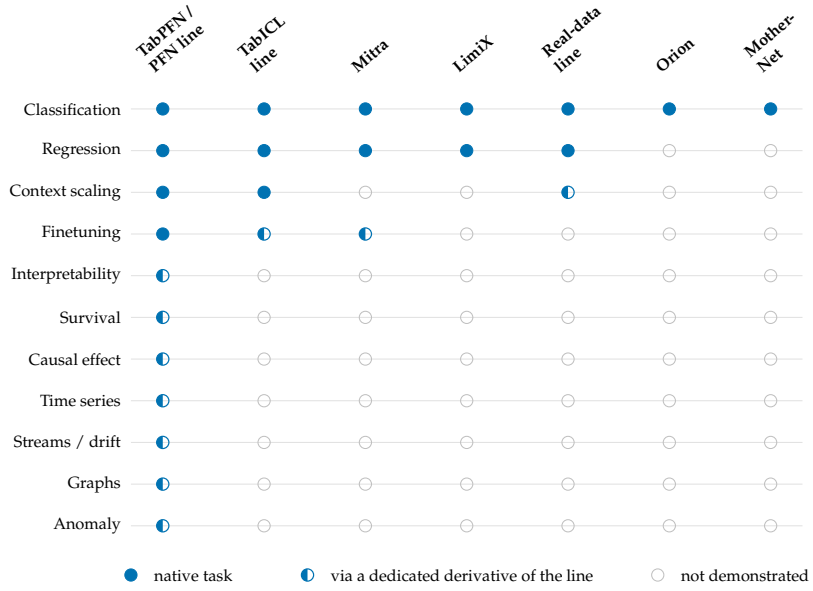


Figure 4. Capability map of the current generation. Rows are capability axes; columns are model lines. A full marker denotes a native task of a model in the line, a half marker a capability demonstrated by a dedicated prior-data fitted derivative of the line rather than the base model, and an empty marker a capability not demonstrated. A line's cell is filled if any version in the line demonstrates the capability. The survival, causal, time-series, stream, graph, anomaly, interpretability and finetuning derivatives are the works cited in Sections 5.1, 5.3, 5.4 and 5.5; the concentration of half markers in the TabPFN column shows that capability breadth beyond classification and regression is, so far, mostly an extension of the synthetic-prior line.

Table 4 sets out, for each of the five axes, the mechanism the capability rests on and the specific limitation that Section 8 weighs against it, so that a strength recorded here can be traced to its independent verdict there.

Table 4. The five capability axes of Section 5, the mechanism each rests on, representative models, and the limitation Section 8 measures against it. The table structures the capability map by axis.

Capability Axis	Mechanism	Representative Work	Key Limitation (Section 8)
Context construction and scaling	Compression, retrieval or re-engineered attention	TuneTables, MixturePFN, TabDPT, TabFlex, Chunked TabPFN [27,59–62]	A wider window is not a wider capability; reported envelopes are vendor-stated
Mechanics and inductive bias	Attention-weighted vote or nearest-prototype readout	TabPFNv2, TabICLv2, Mitra, ModernNCA [63,64]	Mechanism-specific fragilities under engineered stress and shift
Task and output breadth	Re-encoding a task as a table the model already reads	Survival, causal, streaming, graph and anomaly derivatives [65–69]	Mostly single-benchmark; external validity unproven
Interpretability	Post hoc kernel reading or an additive constraint	KernelICL, GAMformer [70,71]	Accuracy and auditability trade against each other
Adaptation	Full finetuning sharpens the retrieval-like readout	TabPFNv2 finetuning study, TabForestPFN [72,73]	Gain is uneven and least where the paradigm is weakest

5.1. Context Construction and Scaling

In-context prediction pays for its convenience with its context. The labelled training set is consumed together with the queries in a single forward pass, so the size of that set is bounded by what the attention mechanism can hold, and the cost of the pass grows with the rows and features it must attend over. The original TabPFN made the constraint concrete, remaining practical for roughly a thousand rows and a hundred features, beyond which the forward pass itself became the binding limit [15]. Context construction, meaning how the training set is selected, compressed or assembled before it enters the model, and context scaling, meaning how far the window can be pushed, are therefore the first capability axis, and the one against which the operating envelopes reported in Section 4 are most directly read.

Three families of answer have emerged. The first compresses or optimises the context instead of enlarging it: TuneTables learns a compact set of context tokens that stands in for a much larger training set, lifting prior-data fitted networks past the size TabPFN could hold [59], and MixturePFN routes queries to a mixture of in-context prompters so that no single pass carries the whole table [60]. The second assembles the context by retrieval, selecting for each query the training rows most relevant to it: TabDPT builds its context this way at pretraining time (Section 4.2), and retrieval-augmented in-context learning extends the idea to language-model backbones, a strand taken up in Section 6 [74]. The third re-engineers the attention itself: TabFlex substitutes a linear-attention formulation for quadratic attention and reaches tables of millions of rows [61], TabPFN-Unleashed improves the inference path for large classification problems [75], and Chunked TabPFN provides exact long-context inference by chunking the attention without retraining [62].

Enlarging the window is not the same as enlarging the capability. The chunked-attention study is instructive because it holds the model fixed and varies only the context length, and it separates three regimes: an early plateau where added rows stop helping, a range of continuous gain, and a regime of degradation where more context makes predictions worse [62]. Compression and retrieval carry a cost of their own, since a context that stands in for the training set is a lossy summary of it, and the choice of

what to keep becomes part of the model. The envelopes reported for the generation, a million rows for TabPFN-3 and fifty thousand rows by one hundred features in under ten seconds for TabICLv2, are the developers' stated operating ranges and not independently measured limits (Section 4.5); whether accuracy and calibration hold across them is a question Section 8 returns to. Context length is thus the first axis on which a headline capability and its independent limit must be read together.

5.2. Mechanics and Inductive Bias

The mechanics of the forward pass are the second axis, and the one on which models that agree on the leaderboard diverge most. Several current architectures reach comparable benchmark accuracy while differing in construction, so accuracy alone cannot say whether they compute the same function or where each will fail. A mechanistic audit of three released classifiers, the per-cell attention of TabPFNv2, the row-token attention of TabICLv2 and the label-slot design of Mitra, put through one evaluation protocol, finds that they do not [63]. TabPFNv2 and Mitra form their prediction as an attention-weighted vote over the labels of the context rows at a late layer, whereas TabICLv2 classifies by the nearest class-mean prototype in a representation built early in its column embedder, its deep in-context trunk adding little. The readouts are confirmed by intervention and are specific to their backbone: transplanting one model's readout onto another collapses accuracy by 33 to 40 percentage points, so readout and backbone are designed together, not assembled from interchangeable parts [63].

The symmetries a tabular model should respect, invariance to the order of rows, of columns and of class labels, are learned by the current models rather than built into them, with one exception. None of the three architectures enforces column or class invariance by construction, yet the pretrained models have largely learned to ignore these orderings, and small edits make the residual invariance exact at no measured cost to accuracy: zeroing a positional weight matrix in TabPFNv2, removing the rotary feature encoding in TabICLv2, and wrapping any of them in a one-versus-all scheme for class order [63]. Mitra is column-invariant through its shared cell embedding and its absence of positional encoding, the one model whose invariance is architectural. The same concern motivates designs that build the symmetry in from the start: EquiTabPFN makes the network equivariant to the ordering of the target components and thereby lifts the fixed-target-dimension constraint of the TabPFN line, matching or improving accuracy on tasks with more classes than were seen in pretraining [76].

The inductive bias that delivers the accuracy is also what governs the failure. Perturbations engineered against each readout, and invisible to a sensible learner, reproduce the predicted breakages: flipping the labels of the most-attended context points degrades the vote-based models, and replacing feature values by their per-column ranks degrades the distance-calibrated readouts of TabPFNv2 and TabICLv2 while leaving Mitra, whose quantile front-end discards absolute scale, untouched [63]. The audit also finds that much of each backbone carries little load, so a large share of the parameters can be removed without hurting accuracy. Whether these mechanism-specific fragilities matter under the feature shift and distribution change of deployment, and beyond engineered stress, is a question Section 8 takes up. Adaptation reshapes these mechanics without replacing them, and how finetuning acts on the same retrieval-like readout is the subject of Section 5.5.

Figure 5 contrasts the two readout archetypes and shows the accuracy that is lost when one is moved onto the other backbone. Models that rank together on a leaderboard need not compute the same function, and the divergence sits in the readout, which is why a benchmark score is a weak guide to how any one line behaves off its tested regime.

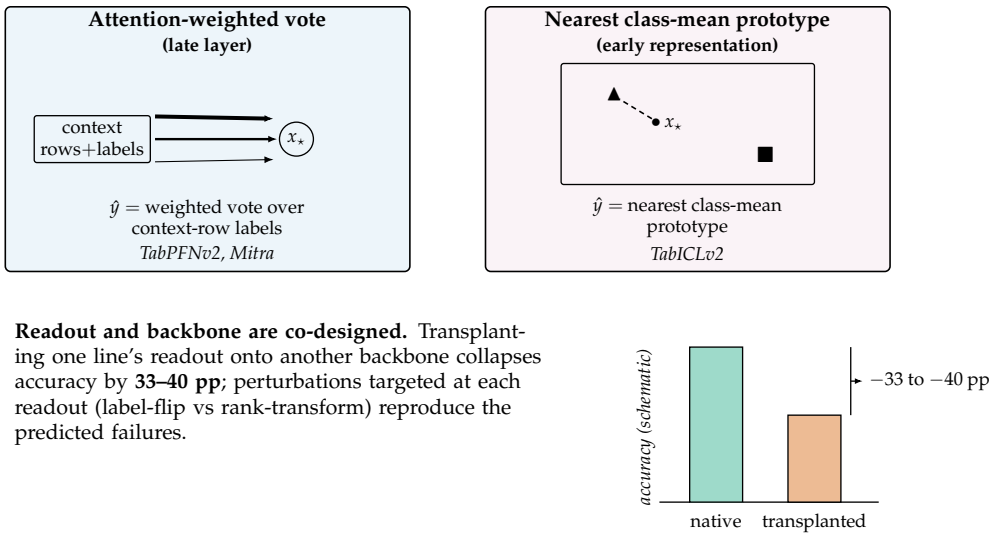


Figure 5. Two readout archetypes identified by the mechanistic audit. TabPFNv2 and Mitra form the prediction as an attention-weighted vote over context-row labels at a late layer, whereas TabICLv2 classifies by the nearest class-mean prototype in a representation built early in its column embedder. Transplanting one line's readout onto another backbone collapses accuracy by 33 to 40 percentage points, so readout and backbone are designed together. The accuracy bars are schematic. In the diagrams, $x_\star$ denotes the query row, the horizontal arrows in the left-hand panel carry the attention weights from the context rows to that query, and the dashed arrow in the right-hand panel marks the distance from the query to the nearer class-mean prototype, the triangle and the square denoting the two class means.

5.3. Task and Output-Modality Breadth

The third axis asks how far the mechanism carries beyond the task it was built for. The paradigm was designed for independent and identically distributed tabular classification and regression, and much of the past year's work has re-encoded other task structures as tables the pretrained model already reads so that the same in-context forward pass serves them without a new architecture. The list is now long. Survival analysis [65], causal effect estimation [66], prediction under data streams and concept drift [67,77], and graph node-level tasks reached by converting a graph to a table [68,78,79] have all been carried onto the in-context mechanism, alongside detection tasks: graph anomaly detection [69], and zero-shot outlier detection, where FoMo-0D predicts inlier or outlier labels for a new table with no fitting and no labelled data [80]. Domain instances such as knowledge tracing extend the same move to sequential educational data [81].

Output modality is the second half of the axis. The models were built to emit a posterior predictive distribution over a target, and the breadth above mostly reuses that output while changing the input encoding. Time-series forecasting is the instructive case: a tabular model can be applied to forecasting by re-encoding the series as a table of lagged features [82], which is a different object from the dedicated time-series foundation models that a companion review treats, and this review keeps the two apart.

The breadth is real but shallow. Most of these results are single-benchmark demonstrations that carry the paradigm's inductive bias into a domain with its own distribution-shift structure, and whether the transferred capability survives outside the demonstrating benchmark is the external-validity question Section 8 takes up. The axis records where the mechanism has reached, and not yet how well it holds there.

5.4. Interpretability

Interpretability is the fourth axis, and the current generation approaches it from two directions, one reading the mechanism after the fact and one constraining the architecture before it. The first rests on the observation that the in-context forward pass behaves like a weighted average over the labels of the context examples, the reading Section 5.2 traced to concrete vote and prototype readouts. KernelICL makes the average explicit, replacing the opaque prediction head with a stated kernel, Gaussian, dot-product or nearest-neighbour so that every prediction is a transparent weighted sum of context labels whose weights can be inspected; on the TALENT suite it holds the accuracy of the unconstrained models while unifying kernel methods, neighbour rules and attention under one taxonomy [70]. Interpretability here is recovered rather than designed in, and it is bought by fixing the readout the audited models had already learned.

The second direction constrains the model to an interpretable functional form from the start. GAMformer estimates the shape functions of a generalised additive model in a single in-context forward pass, so the fitted model is additive by construction and each feature's contribution can be read off, at accuracy comparable to iteratively fitted additive models [71]. The additive tabular transformers introduced in Section 4.4 impose the same generalised-additive constraint through the architecture [58]. Both routes give up the unconstrained flexibility that supplies the headline models their accuracy for a form a practitioner can audit.

The trade is the difficulty. A post hoc kernel reading is faithful only to the extent that the fixed readout captures what the full model does, and an additive constraint forgoes the feature interactions an unconstrained transformer can represent, so interpretability on this axis is presently available at the cost of the mechanism or at the cost of flexibility. Where a decision-facing deployment needs both an inspectable prediction and the full accuracy of the unconstrained models, the axis is unresolved, and it enters the research agenda of Section 8.

5.5. Adaptation: Few-Shot Learning, Transfer and Finetuning

Frozen in-context prediction is the paradigm's default, but the same models accept gradient-based adaptation, and the question is how far adapting the weights helps and when the frozen forward pass is already enough. Earlier work reached opposing recommendations, some advising partial or parameter-efficient finetuning to avoid overfitting the small target sets on which the models are strongest. A systematic study of TabPFNv2 revisits the choice and finds that full finetuning, properly configured, is the most practical option, matching or beating low-rank and last-layer alternatives while converging faster [72]. The same study reads the effect mechanistically and connects it to the retrieval-like readout of Section 5.2: finetuning sharpens the query–key dot products in the model's attention so that they track target similarity more closely, and the gains concentrate on the query points where attention becomes more sharply focused, improving the implicit retrieval the frozen model already performs [72]. Adaptation refines the existing mechanism and does not install a new one.

Around this core sit complementary adaptation strands. Retrieval-and-finetuning assembles a query-specific context and then adapts the model to it [83]; few-shot augmentation [84] and variance-aware initialisation [85] target the small-sample regime; and hybridisation with gradient-boosted trees carries the pretrained representation into an ensemble instead of discarding it [86].

The benefit is uneven, and the unevenness is the point. Finetuning helps more as the target grows; on the small datasets where in-context prediction is already strongest, its improvement over the frozen model is often not statistically significant [72]. Within

the size a single large-memory GPU can hold, roughly a million table cells, finetuned TabPFNv2 reaches the best reported results on academic benchmarks; on datasets with gradual temporal shift and rich engineered features it is less stable, and non-foundational deep models remain ahead [72,87–90]. Adaptation thus buys accuracy where the paradigm is already strong and buys least where it is weakest, so the open costs are scalability beyond the single-GPU context and stability under the real-world shift that Section 8 examines.

6. The LLM-on-Tabular Bridge

A parallel line of work builds no dedicated tabular model at all, and instead adapts a general-purpose language model to work over tables by rendering their contents as text. The survey of Fang et al. [34] mapped this landscape broadly, organised by task, technique and metric, before the release train of Sections 4 and 5 took shape. This section isolates the three strands of that landscape that bear on tabular prediction, prediction from serialised rows, language-model-driven feature engineering, and language-model tabular generation, and reads each against the dedicated foundation models of the preceding sections. The scope stated in Section 1 continues to bind the reading: table question answering, NL2SQL and table reasoning through symbolic execution concern a different modality of work over tables and are excluded, so systems whose contribution is execution binding or general table agency are set aside, and only prediction, feature engineering and generation are treated here.

What unites the strand is serialisation, the conversion of a row's columns and values into a natural-language string a pretrained model can read, and serialisation is the source of both its reach and its limits. It lets a language model use the meaning carried by column names and category labels, the information the dedicated models of Section 4 discard when they treat a table as an anonymous array of numbers and category codes, and it makes zero-shot and few-shot prediction possible from feature names alone. The same conversion spends a token budget on every value, handles continuous quantities coarsely, and replaces a single amortised forward pass with autoregressive decoding, so the approach is strongest where labelled data is scarce and semantics are rich, and weakest as tables grow numeric and large. Section 8 tests whether the reported advantages survive independent evaluation.

Table 5 summarises the three strands, the representative methods behind each, and the limitation each carries into the assessment of Section 8. The subsections that follow take the strands in turn.

Table 5. The three strands of the language-model-on-tabular bridge, their representative methods and the limitation each carries. Systems whose contribution is table question answering, NL2SQL, symbolic-execution table reasoning or general table agency (for example Binder and Table 2) are excluded under the scope stated in Section 1.

Strand	What the Language Model Does	Representative Methods	Recurring Limitation
Prediction via serialisation (Section 6.1)	Reads a serialised row and returns the label, prompted or fine-tuned	TabLLM, LIFT, UniPredict, Tabula-8B, PTab; efficiency-driven variants Latte, frozen-representation and retrieval methods, insight distillation, and anomaly detection [29,91–99]	Long prompts, coarse numeric handling and decoding cost as sample size grows and columns become numeric
Feature engineering (Section 6.2)	Proposes feature transformations a downstream predictor then consumes	CAAFE, decision-tree-guided generation, LLM-guided AutoML, external-knowledge benchmarking [30,100–102]	Bounded by how well the schema describes the data and by the downstream model's capacity

Table 5. *Cont.*

Strand	What the Language Model Does	Representative Methods	Recurring Limitation
Generation (Section 6.3)	Models a serialised row and samples synthetic tables	GReaT, TabuLa, CATS, TAGAL, SampleLLM, LLM–diffusion hybrids, differentially private and explainability-guided and augmentation variants [31,103–112]	Fidelity on numeric columns and correlations, privacy leakage, and the cost of decoding long sequences

6.1. Prediction via Serialisation

Serialisation-based prediction turns each row into a textual description and asks a language model to read off the label, so the meaning in schema and category names becomes usable evidence. TabLLM established the recipe for the few-shot regime, serialising rows to text and adapting a language model to classify them, and its advantage over conventional baselines is largest when only a handful of labelled examples are available [29]. The appeal is a model that predicts from the description of a task as well as from its numbers.

Around this recipe, a family of designs varies the interface and the scale. One line adapts a language model through a natural-language interface so that a single model serves many tabular tasks [91], and a related line instruction-tunes one predictor across tasks [92]; a further line pretrains a large tabular language model on billions of serialised rows for zero-shot and few-shot use [93], and an earlier line encodes rows textually and models them with a masked language objective [94]. A second, efficiency-driven line keeps the prompt small: it transfers a language model's latent knowledge for few-shot learning [95], uses frozen language-model representations as compact features [96], retrieves only the most relevant rows into the prompt [97], or distils data-driven insights from the training set before classifying [98]. The two lines answer the same problem from opposite ends, one enlarging the model's tabular competence and the other lowering the cost of using it, and the strand has been carried to anomaly detection as well as to classification and regression [99].

The difficulty is that the advantage is narrow. Serialisation is strongest in the semantics-rich, very-few-shot setting and fades as the sample size grows and the columns become numeric, where a serialised prompt is long, the numeric handling is coarse, and autoregressive decoding is far more expensive than the dedicated models' single pass. Where the serialisation advantage actually holds against dedicated tabular models, and at what cost, is a question Section 8 returns to.

6.2. Feature Engineering and Data-Science Agents

Feature engineering is where domain knowledge has always entered tabular modelling, and it is mostly manual, which makes it a natural target for a model that already holds world knowledge and can read a schema. In this strand the language model does not predict; it proposes the inputs a downstream predictor, a gradient-boosted ensemble or a tabular foundation model, then consumes. CAAFE is the reference instance: given a dataset and its textual description, a language model proposes feature transformations, and each is retained only if it improves validation performance, so the proposals are filtered by evidence rather than accepted on trust [30].

Later work steers the proposals and tests their worth. Decision-tree reasoning guides which transformations a language model generates [100]; language-model-guided feature generation with selective ensembling has been folded into automated anomaly-detection pipelines [101]; and a benchmark of externally supplied domain knowledge measures when such assistance helps a data-science agent and when it misleads, reporting that the benefit is uneven and depends on the quality of the knowledge supplied [102]. The strand is bounded on two sides, by how well the schema describes the data and by the capacity

of the downstream model that consumes the engineered features, and its gains are real where both are favourable and thin where either is not. Section 8 reads these claims against independent evidence.

6.3. Tabular Data Generation

Synthetic tabular data is wanted for augmentation, for privacy-preserving sharing and for balancing rare classes; the classical generators (generative adversarial networks, diffusion models and copulas) reproduce mixed types and column semantics awkwardly. An autoregressive language model instead models a serialised row directly and learns the joint distribution over its fields. GReaT demonstrated this route, generating realistic synthetic rows with an autoregressive language model and conditioning flexibly on any subset of columns [31].

The strand has since branched along fidelity, efficiency and privacy. Efficiency-oriented synthesis lowers the training and sampling cost of the language-model generator [103,106]; correlation augmentation [104] and agentic generation [105] target the multivariate structure the base method reproduces poorly; and a hybrid route couples a language model with a diffusion model to combine semantic and numeric fidelity [107]. Augmentation-focused variants generate additional rows and features for small datasets [111,112], a privacy line adapts language-model generators to satisfy differential privacy [108,109], and an explainability-guided variant uses feature attribution to steer what is generated [110]. The volume of this work has been large enough to attract its own surveys [113,114].

Generation is the least settled part of the bridge. Fidelity on continuous columns and on multivariate correlations, the privacy actually delivered against membership and reconstruction attacks, and the cost of decoding long serialised sequences are open on every generator in the strand. A health-data benchmark that reads several of them against utility and privacy together makes the trade-offs concrete [115]. These quality gaps are examined in Section 8.5, which this section does not pre-empt.

6.4. Language Models and Dedicated TFMs Compared

The two families answer different questions well. Serialisation pays where a table's meaning lives in its names and text, where labels number in the tens or fewer, and where a task rewards the world knowledge a language model already holds; it also travels naturally across heterogeneous schemas since text is a common representation. Dedicated tabular foundation models dominate where tables are numeric and larger, where a calibrated probability is needed, and where serving latency matters because a single amortised forward pass is cheaper than token-by-token decoding and the models of Section 4 are built for that regime.

The relationship is complementary. The language-model strand supplies the semantic channel the dedicated models discard, and the dedicated models supply the calibrated, low-latency prediction the language-model strand struggles to match, so the pragmatic designs sit between them: a language model engineers features or assembles a serialised context that a tabular foundation model then predicts from, a hybridisation whose deployment side is taken up in Section 7.3. Whether the serialisation advantage survives independent evaluation, and whether language-model generation is fit for use, are the reality checks Section 8 carries out, so the positioning offered here is a map of where each family should be expected to win, to be confirmed against the evidence there.

7. Applications and Deployment Considerations

This review is organised by capabilities and architectures, and it treats applications and deployment briefly. The applied depth, meaning industrial predictive maintenance,

sensor pipelines and domain-specific deployments, belongs to a companion review and is out of scope here. The purpose of this section is to record that the generation has left the benchmark for applied use, to describe how it changes the shape of applied work, and to name the two practical concerns that shape that use: how the models are served, and how open they are.

7.1. The Shift in the Practitioner Workflow

Before the applications themselves, it is worth stating what the paradigm changes in the daily practice of applied machine learning because the architectural account of Section 4 does not make this visible. The classical tabular workflow runs through five stages: the problem is formulated, the data are cleaned and features are built, a learning algorithm is selected from a candidate set, its hyperparameters are searched, and the fitted model is validated and deployed. The middle two stages consume the bulk of the compute budget and much of the practitioner's time, and they are the stages the amortised forward pass absorbs. A pretrained in-context model arrives with its architecture and its inductive bias already fixed by the prior, and it produces a prediction without a training run or a tuning loop, so on the data regime where it is strongest the model-search question largely disappears (Figure 6).

The consequence is a redistribution instead of a net saving. When model search stops dominating the effort, what remains is the work the paradigm does not perform: deciding what is being predicted and for whom, assembling and cleaning the table, choosing and encoding the features, and deciding which rows constitute the context. That last item is new and specific to the paradigm because the labelled context is an input to the forward pass, so its size, its class balance and its representativeness become design choices that carry the weight hyperparameters once carried, and the evidence of Section 5.1 shows that accuracy is sensitive to them. Feature quality matters as much as before since the serialisation and semantics-aware lines read column names as meaning (Sections 4.3 and 6.1).

A second consequence is an obligation rather than a saving. The stages that disappear are also the stages in which a practitioner learned the behaviour of the model on the data: a hyperparameter sweep produces a validation curve, and a model comparison produces a sense of variance. The single forward pass produces none of this, and the independent evidence of Section 8 shows that the properties a decision-facing deployment needs, calibrated coverage and group fairness among them, are not supplied by the pass. Validation therefore acquires an explicit reliability step that the classical workflow could leave implicit. The net effect is that the paradigm lowers the barrier to a first working model and raises the relative weight of problem formulation, data quality and reliability checking, which is a different distribution of expertise and not a smaller one.

7.2. Representative Applications

The reach is already cross-domain. In cybersecurity, TabPFN and TabICL have been evaluated for tabular intrusion detection [116]; in process engineering, tabular foundation models have been benchmarked for the prediction of total volatile fatty acids in anaerobic digestion [117]; and in education, the in-context mechanism has been carried to live knowledge tracing [81]. These instances share a shape: a small-to-medium tabular target in a domain where per-dataset training was the norm, met by a pretrained model that predicts without a training run. The depth of any one domain, its data pipeline, its operating constraints and its regulatory setting, is the subject of applied studies and is not pursued here.

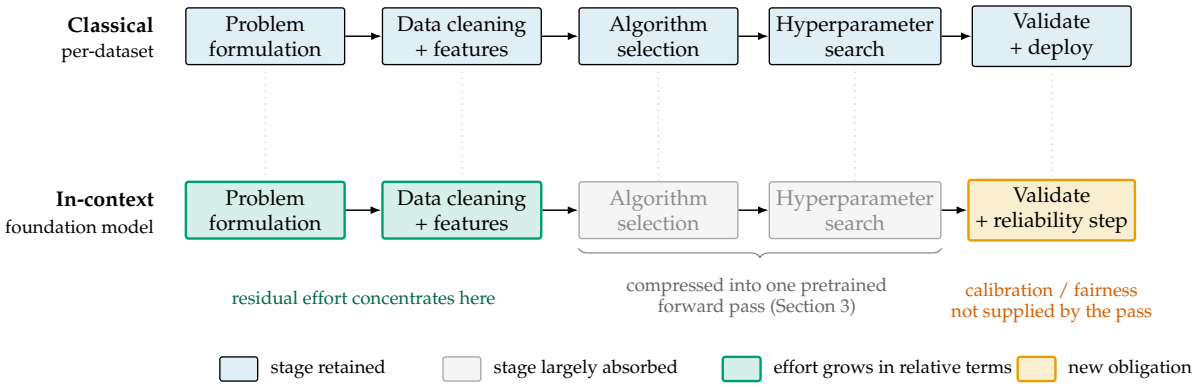


Figure 6. Where practitioner effort goes in the classical tabular workflow and under an in-context tabular foundation model. Algorithm selection and hyperparameter search are absorbed into the pretrained forward pass on the regime where the paradigm is strongest, so the residual effort concentrates on problem formulation, data quality and feature and context construction, while validation acquires an explicit reliability step that the forward pass does not supply (Sections 5.1 and 8). The horizontal arrows give the order of the stages within each workflow, the vertical dotted lines link each classical stage to its counterpart under the in-context model, and the brace marks the two stages absorbed into the pretrained forward pass.

7.3. Serving, Distillation and Tooling

Deployment turns on the cost of the forward pass. In-context prediction reads the whole training set at inference, so serving latency and memory grow with the context, and two responses have emerged. The first carries the pretrained representation into a cheaper model, hybridising a tabular foundation model with gradient-boosted trees so that the ensemble inherits the pretraining without paying the full in-context cost at serving time [86]. The second standardises the engineering: a unified library provides a common interface for inference and finetuning across the model families, which lowers the cost of adopting and comparing them [118]. Both address the same practical limit, that a single amortised forward pass is convenient to train against and not always cheap to serve.

Table 6 organises the intrinsic side of this cost by the technique that governs it. The per-query cost scales with the context a model holds, and the routes that manage it, factorised and sparse attention, learned context compression, retrieval of only the relevant rows, and hypernetwork emission of a small serving network, each buy scale by giving something up: an approximation of full attention, a lossy summary of the training set, a per-query selection step, or a single pass that consumes the context once. The convenience of a single amortised forward pass is therefore real but priced, and that price is the axis on which deployment decisions turn.

7.4. Openness and Licensing

Openness varies sharply across the generation, and it constrains both independent evaluation and applied adoption. Table 7 records, for each model, the availability of weights, the licence, the availability of code, and the disclosure of pretraining data. The permissive end is well populated: TabICL and its second version are released under BSD-3-Clause, and Mitra and TabDPT under Apache-2.0, with open weights and code. The restrictive end is occupied by the most capable models, since the weights of TabPFN-2.5 and TabPFN-3 are released for research and internal evaluation under non-commercial terms. Two cases sit between the ends and matter for reproducibility. LimiX releases its code under Apache-2.0 while its weights are academic-open and commercial only by authorisation; and ConTextTab, released as SAP-RPT-1-OSS, carries Apache-2.0 code while its checkpoints inherit the research-only terms of the corpus they were trained on. The consequence is

that a study which reproduces a headline result on the permissive models cannot always reproduce it on the restrictive ones under the same terms, so openness is itself a variable in the independent evaluation Section 8 calls for.

Table 6. The compute cost of in-context prediction, organised by the technique that governs it. In-context models re-read the training context at every prediction, so the per-query cost scales with the context that is held; the entries below are the routes by which the generation manages that cost, with the trade each carries.

Cost Regime or Technique	Representative Models	Effect on the Per-Query Cost	Trade
Dense in-context attention	TabPFN v2, Mitra [17,22]	Cost grows quadratically in context rows in one forward pass	Bounds the practical table size to small and medium
Factorised attention	TabICL, TabICLv2 [20,21]	Column embedding then row-wise attention lowers the cost of large tables	Reported $5\times10^4\times100$ in under ten seconds on one GPU
Linear or sparse attention	TabFlex, Orion-MSP [24,61]	Sub-quadratic attention reaches millions of rows or beyond one hundred features	Approximation of full attention
Context compression	TuneTables, MixturePFN [59,60]	A compact learned context stands in for a larger training set	The compressed context is a lossy summary
Retrieval-assembled context	TabDPT, TabR, retrieval-ICL [27,74,119]	Only the rows relevant to each query enter the pass	Adds a per-query selection step
Hypernetwork emission	MotherNet, HyperFast [54,120]	Emits a small child network, so serving is decoupled from the context	The context is consumed once, not at every query

Table 7. Openness and licensing of the current generation. Weights and licence facts were taken from each model's official card or repository and verified at source on 13–14 July 2026; n/r = not reported, n/v = not verified at source at the time of writing. "Synthetic" pretraining denotes prior-generated data with no real corpus.

Model	Weights	Licence	Code	Pretraining Data
TabPFN v2 [17]	open	Prior Labs licence (Apache-2.0 with attribution)	open	synthetic (SCM priors)
TabPFN-2.5 [18]	open, gated	non-commercial (research and internal evaluation)	open	synthetic
TabPFN-3 [19]	open, gated	non-commercial	open	synthetic
TabICL [20]	open	BSD-3-Clause	open	synthetic
TabICLv2 [21]	open	BSD-3-Clause	open	synthetic
Mitra [22]	open	Apache-2.0	open	synthetic (SCM and tree priors)
LimiX [23]	academic-open, commercial by authorisation	code Apache-2.0; weights academic terms	open	synthetic (hierarchical SCM)
TabFM [25]	open	code Apache-2.0; weights TabFM Non-Commercial License v1.0	open	synthetic (SCM priors)
TabDPT [27]	open	Apache-2.0	open	real tables

Table 7. *Cont.*

Model	Weights	Licence	Code	Pretraining Data
Real-TabPFN [28]	n/v	n/v	n/v	synthetic, continued on real corpora
ConTextTab (SAP-RPT-1-OSS) [26]	research-only checkpoints	code Apache-2.0; checkpoints inherit corpus terms	open	real corpus (T4 subset of TabLib)
MotherNet [54]	research prototype	research-only	open (microsoft/ticl)	synthetic
Orion-MSP [24]	n/r	MIT (announced)	announced	synthetic
Orion-BiX [55]	n/r	MIT (announced)	announced	synthetic (causal priors)

7.5. A Practitioner Decision Guide

Figure 7 and Table 8 turn the evidence of the preceding sections into a working rule for practitioners. The paradigm is the reasonable default on small, independent and identically distributed tables, and on semantics-rich data with very few labels; large, high-dimensional or shifted data still call for a tuned gradient-boosted or deep model; and a requirement for calibrated uncertainty or fairness calls for an explicit step on top of whichever base model is chosen. The guide is deliberately conservative, with its cut points taken from the independent evidence that the rest of the review assembles.

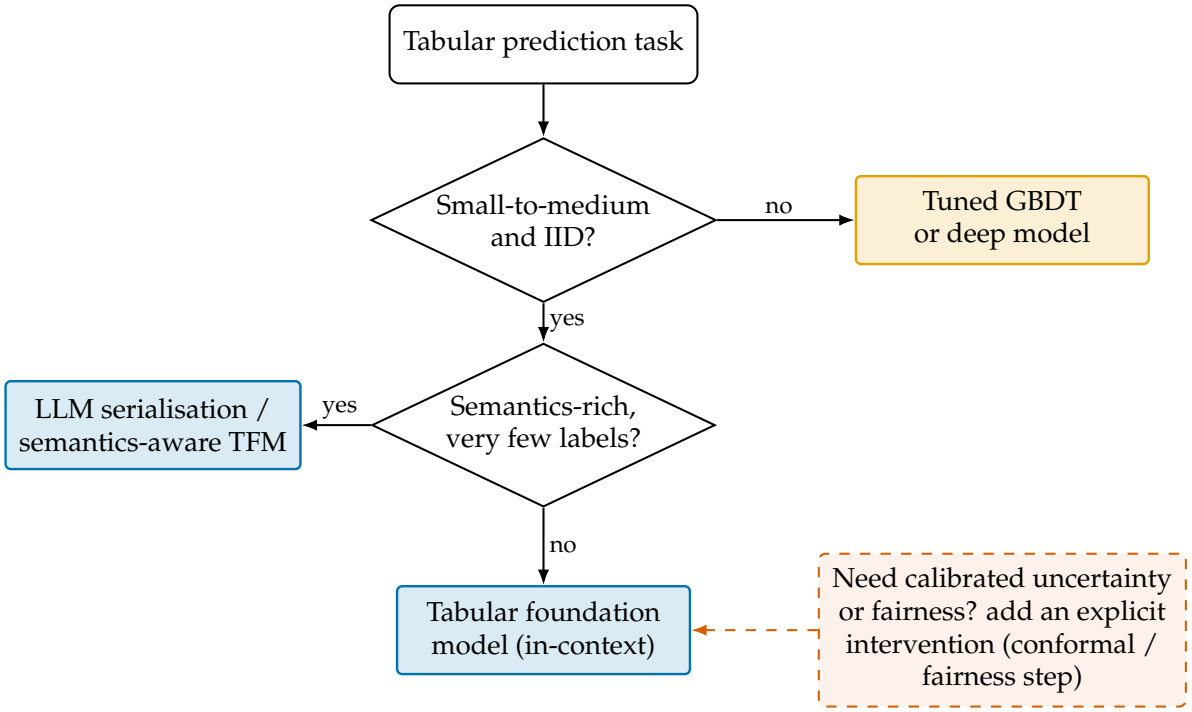


Figure 7. A decision guide for choosing an approach to a tabular task, synthesised from Sections 5 and 8. Small independent and identically distributed tables favour a foundation model; large, high-dimensional or non-IID data favour a tuned gradient-boosted or deep model; a requirement for calibrated uncertainty or fairness calls for an explicit intervention. The arrows carry the flow of the decision, each arrow leaving a diamond is labelled with the answer, yes or no, that selects it, and the dashed arrow marks the intervention added on top of whichever base model is chosen.

Table 8. A decision guide from the evidence of Sections 5 and 8, stating the regime in which each approach is the reasonable default, the caveat that qualifies it, and the practical consideration that governs putting it into service; the tabular counterpart of Figure 7. GPU = graphics processing unit.

Data Regime	Reasonable Default	Rationale	Caveat	Deployment Consideration
Small-to-medium, IID, numeric	Tabular foundation model, in context	Best reported results on small IID data with no tuning	Confirm calibration and fairness separately	Context ships with the model at inference; check the weights licence before commercial use (Table 7)
Semantics-rich names or free text, very few labels	Language-model serialisation, or a semantics-aware model (ConTextTab, TabSTAR)	Uses the meaning in column names; strong in the very-few-shot setting	Fades as the data grow numeric and large; decoding cost	Token cost scales with rows and columns; an external interface may place data outside the trust boundary
Large ($\gtrsim 10^5$ rows) or high-dimensional	Tuned gradient-boosted or deep model	Tree-based and deep models remain ahead in this regime	Foundation-model context and latency cost grow	Training cost is paid once and inference is cheap, which suits high-throughput serving
Non-IID, temporal or grouped shift	Tuned gradient-boosted or deep model with shift-aware validation	Foundation-model accuracy cedes under distribution shift	Feature-shift robustness is unresolved	Budget for periodic refitting and shift monitoring; time-based splits at validation
A coverage guarantee is needed	Any base model with an explicit conformal step	The emitted posterior is less well covered than that of trees	The forward pass alone does not supply the guarantee	Reserve a calibration split; the conformal step adds a second inference stage
A fairness guarantee is needed	Any base model with an explicit fairness intervention	Fairness is not a by-product of the prior or the backbone	The base model does not perform the intervention	Sensitive attributes must be available at calibration time and governed under the applicable policy
Latency-critical or offline serving	Distilled or tree-based model	A per-query transformer pass over the full context can be too slow	Distillation forfeits part of the in-context advantage	Distillation to a compact model is the documented route (Section 7.3); GPU availability governs the alternative

8. Critical Assessment: Limits, Failure Modes and Open Problems

Section 5 recorded the generation's capabilities as claims and stated that each would be read again here against independent evaluation. This section carries out that reading. The capabilities the developers report are measured mostly on curated benchmarks that the model-developing laboratories helped to define, so the question is not whether the reported numbers are correct but whether they survive when the evaluation moves to harder data regimes, to reliability in place of accuracy, and to properties the benchmarks do not score. The evidence assembled below is recent and, in the two anchor cases, comes from within the same research community that builds the models, which makes it a calibration of the paradigm and not a case against it. Table 9 sets the reported capabilities of Section 5 against the independent findings, one row per axis; the subsections take each in turn, and Section 8.9 turns the open ends into a research agenda.

Figure 8 reads the two columns side by side: each capability is present as a reported strength, whereas the independent verdict is established only for accuracy on curated data and settles at partial or open across generalisation, calibration, feature-shift robustness, fairness and generation. That asymmetry is the thesis of this section, and it fixes the order in which the subsections take up each axis.

Table 9. The reported capabilities of Section 5 set against the independent findings synthesised in this section. AUC = area under the ROC curve; GBDT = gradient-boosted decision tree; IID = independent and identically distributed.

Reported Capability (Section 5)	Independent Finding	Study
Best reported accuracy on curated IID benchmarks	Strongest on tiny-to-medium IID data; in these studies tree-based and deep models lead on non-IID, large and high-dimensional data	[121–123]
Small-data classification gains	Hold against carefully tuned baselines, not against leaderboard margins taken at face value	[124–126]
Deep and pretrained models now rival tuned trees on many tasks	True on average across 300+ datasets, yet gradient-boosted trees stay very strong and the best results concentrate in a few models	[127,128]
Wins reported on curated suites	The lead narrows sharply once the model is evaluated in open, shifted environments	[129]
Progress read from public tabular repositories	Unreflected repository use can inflate it; benchmark provenance and curation matter	[130,131]
Prediction from serialised feature names, few-shot (Section 6.1)	Not a universal predictor; accuracy uneven on small tabular datasets	[132]
Posterior predictive distribution and uncertainty	Highest AUC but lower conditional coverage than GBDTs under conformal prediction	[133,134]
Task breadth on a stable feature space (Section 5.3)	Accuracy degrades under feature shift in open environments	[135]
Handling of missing features	Train-time masking helps only under leakage-controlled conditions	[136]
Fairness from the pretraining prior or the language-model backbone	Group disparities remain; fairness requires an explicit intervention	[137–140]
Language-model tabular generation (Section 6.3)	Trails diffusion generators on multi-table fidelity	[141]

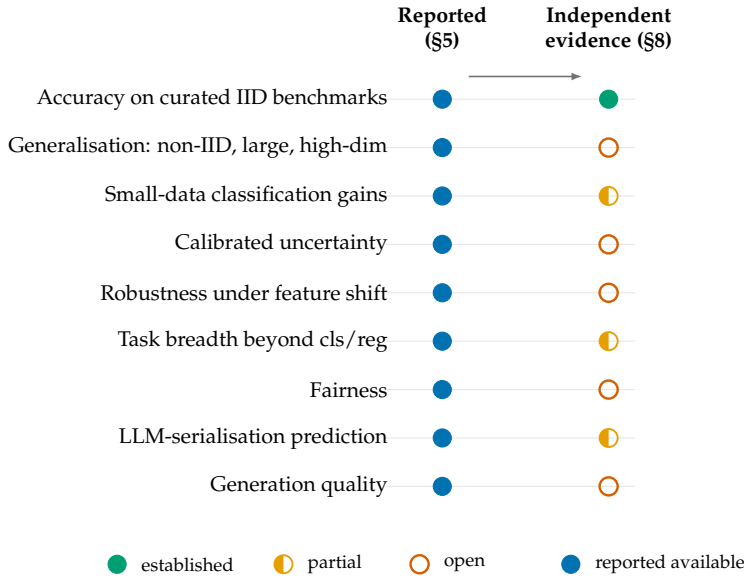


Figure 8. Each capability reported in Section 5 set against the verdict of the independent evidence in Section 8. The verdict is encoded by both marker shape and colour, so the figure reads in greyscale: established (green, filled marker), partial (orange, half-filled marker) and open (vermilion, open marker); the reported-available column of Section 5 uses blue filled markers. A schematic synthesis of Table 9.

8.1. Generalisation Beyond Curated Benchmarks

The headline capability is accuracy on curated benchmarks, and the first question is how far that accuracy travels. BeyondArena provides the most searching answer to date: across eleven models and 142 curated datasets spanning independent and identically distributed (IID), temporal and grouped task types, at sample sizes from tiny to a million rows and over low- and high-dimensional feature spaces, tabular foundation models are strongest on tiny-to-medium IID data, while tree-based and deep models remain ahead on the non-IID, large and high-dimensional datasets [121]. The result carries weight partly because of who reports it: the benchmark is built by the same laboratories that develop the leading models, and its argument is that standard suites are defined where the paradigm already wins, so progress measured on them tracks marginal gains on the easy regime and not the harder one. A concurrent re-benchmark of tabular classification reaches a compatible reading, placing the reported gains of the current models against carefully tuned baselines rather than accepting leaderboard margins at face value [124]. The language-model route to prediction is tested on the same axis and found uneven: on small tabular datasets a general language model is not a universal predictor, and its accuracy trails dedicated methods on the regime where the serialised approach was expected to help [132,142,143]. Read together, these results locate the paradigm's accuracy where Section 5.1 located its mechanism, on small IID tables, and indicate that the reported strength does not yet extend to distribution shift, scale and dimensionality. On the studies reviewed here the capability is established on its home regime and unproven beyond it. The statement is a synthesis of the evidence retained in Section 2 and not a general law: it rests on five studies, the largest of which is itself authored within the model-developing community (Section 8.6), and a benchmark that sampled the non-IID regime differently, or a model line that closed the gap after the search date, could revise it.

Figure 9 compresses these axis-by-axis findings into a single map of where each family leads, on the plane of sample size against feature dimensionality. The favourable region for the paradigm is a bounded corner of tiny-to-medium, low-dimensional, independent and identically distributed data, and under temporal or grouped shift that corner contracts further towards the origin. The boundary of this region, and how far it can be pushed outwards, is the concrete form the generalisation question takes for the rest of the review.

8.2. Calibration and Uncertainty

Accuracy is not reliability, and the second axis asks whether the posterior predictive distribution the models emit can be trusted. A dedicated uncertainty benchmark compares four tabular foundation models with gradient-boosted trees and classical baselines on the 112 datasets of the TALENT suite and finds a performance–reliability trade-off: the foundation models attain the highest area under the curve, yet their conditional coverage under conformal prediction is lower than that of the trees [133]. Complementary synthetic experiments locate the regimes in which the gap widens. The finding cuts against a natural reading of the paradigm, since the models are trained to approximate a Bayesian posterior predictive and might be expected to carry calibrated uncertainty with them; on the evidence, the accuracy lead does not transfer to reliability, and a decision-facing deployment that needs a coverage guarantee cannot yet take one from the forward pass alone. Well-calibrated uncertainty is, on this benchmark, an open requirement for adoption and not a settled property.

Figure 10 places the two families on the accuracy-coverage plane and makes the trade explicit: the paradigm sits high on predictive accuracy and low on conditional coverage, while the corner it would need for decision-facing use, accurate and calibrated at once, stays empty. Accuracy and reliability are, on this evidence, separate axes, so a deployment

that requires a coverage guarantee must supply it by a separate step; the forward pass alone does not provide one.

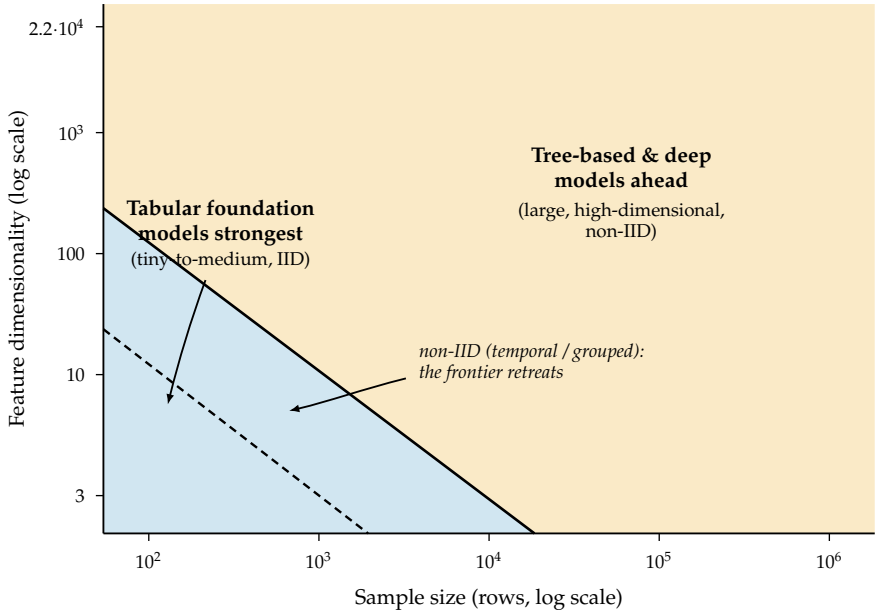


Figure 9. The regimes in which each model family leads, synthesised from BeyondArena [121]. Tabular foundation models are strongest on tiny-to-medium independent and identically distributed (IID) data; tree-based and deep models lead on non-IID, large and high-dimensional data, and under temporal or grouped shift the frontier retreats towards smaller and lower-dimensional data. Axes are logarithmic; the regions are a schematic synthesis of the reported result, not measured points. The solid line is the boundary between the two regions, the dashed line is that boundary under temporal or grouped shift, and the arrow marks the direction in which the frontier retreats.

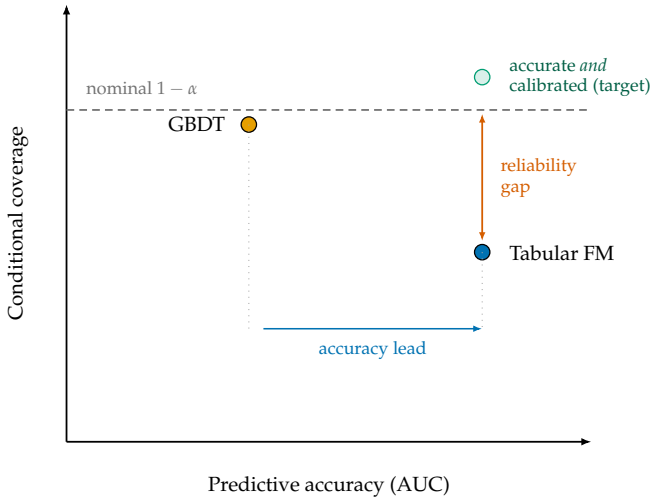


Figure 10. The performance–reliability trade-off reported by the uncertainty benchmark on the 112 datasets of the TALENT suite [133]. Tabular foundation models attain the higher predictive accuracy but a lower conditional coverage than gradient-boosted decision trees under conformal prediction. The positions are schematic.

8.3. Robustness Under Feature Shift and Leakage Controls

The third axis is robustness when the features available at test time are not the features seen at training time. A feature-shift benchmark for open environments shows that tabular classifiers, foundation models among them, lose accuracy as features are removed or drift

away from the training distribution, so the breadth catalogued in Section 5.3 is bounded by an assumption of a stable feature space [135,144,145]. A leakage-controlled audit of the standard remedy, train-time masking of features, refines the picture: masking helps only under specific conditions, and only a protocol that controls for information leakage between the masking and the evaluation separates a genuine gain from an accounting effect [136]. Taken together, the two studies caution that the single amortised forward pass, which treats the query as drawn like the context, does not by itself confer robustness to the feature shift that applied deployments routinely present.

8.4. Fairness

Fairness is a property the benchmarks of Section 5 do not score, and four audits ask whether the paradigm supplies it. The synthetic causal prior that anchors the leading line does not, on inspection, confer it: an empirical study of TabPFN under a fairness lens finds group disparities that the prior does not remove [137], and a critical evaluation on standard algorithmic-fairness benchmarks reaches the same conclusion for its learned representations [138]. The disparities can be reduced by design: a preprocessing method that limits the sensitive information carried by the in-context examples, through latent concept variables, recovers fairness at a marginal accuracy cost [139]. The language-model route offers no free improvement since a direct comparison of large language models with traditional machine learning on tabular classification finds that the language models are not systematically fairer [140]. The theme across the four is consistent: fairness is not a by-product of the pretraining prior or of the language-model backbone, and where it is achieved, it is achieved by an explicit intervention the base model does not perform.

8.5. Generation Quality Gaps

Generation was flagged in Section 6.3 as the least settled part of the bridge, and the independent evidence confirms the caution. A direct comparison of multi-table generation finds a gap between diffusion-based and transformer-based generators, with the transformer route that the language-model strand relies on trailing on the fidelity of the generated tables [141]. The language-model generator's advantages, semantic conditioning and flexible prompting, do not yet close the distance to the diffusion baselines on the multivariate structure that synthetic tabular data must preserve. Generation quality remains the weakest of the capabilities this review has surveyed.

8.6. Benchmark Independence

One structural point qualifies the evidence above. The living benchmark that anchors much of the reported progress, TabArena, is developed by the same community that builds several of the leading models [39], and BeyondArena, the most demanding test of generalisation cited here, extends it through an overlapping group of authors [121]. This is recorded as a fact about provenance and not as an imputation of bias, and the same community produced the most searching critique of the paradigm, which is the healthiest form the situation can take. It nonetheless follows that vendor-reported and leaderboard results are best read alongside evaluations on suites assembled independently, such as the TALENT collection and public OpenML tasks, and that a reported margin on a co-developed benchmark is a weaker claim than the same margin reproduced on an independent one.

Table 10 sets out the benchmarks that certify progress with their scope and their provenance, so that any reported margin can be weighted by where it was measured. The independently assembled suites in the table, among them the TALENT collection, public OpenML, TabZilla, AMLB and TabReD, are the ground on which a co-developed result

is best re-read, and the spread of scopes shows how much of the space, non-IID, large, high-dimensional and temporal, a small-IID leaderboard leaves untested.

Table 10. The benchmarks that certify progress for the current generation, their scope and their developer provenance. Provenance is recorded as a fact and not as an imputation of bias; a margin on a co-developed benchmark is a weaker claim than the same margin reproduced on an independently assembled suite.

Benchmark	Scope (Task Types, Size Range)	Developer Provenance	What It Can and Cannot Certify
TabArena [39]	Living benchmark, 51 curated small-to-medium IID datasets	Built by the community that develops several leading foundation models	Certifies small-IID ranking; does not cover non-IID, large or high-dimensional data
BeyondArena [121]	142 datasets across IID, temporal and grouped tasks; 100 to 10^6 rows; 3 to 2.2×10^4 features; text and high-cardinality columns	Extends TabArena through an overlapping author group	The most demanding generalisation test cited here; still authored within the model-developing community
TALENT [127]	Toolbox of over three hundred datasets; the uncertainty study [133] uses 112	Assembled independently of the model developers	Usable as an independent suite for accuracy and calibration
TabZilla [146]	19 algorithms across 176 datasets; a suite of the 36 hardest	Independent of the leading model developers	An apples-to-apples neural-versus-tree comparison on small-to-medium tasks
AMLB [147]	9 AutoML frameworks across 71 classification and 33 regression tasks	Independent of the leading model developers	An AutoML-framework benchmark, not foundation-model-specific
TabRepo [148]	A large repository of precomputed tabular model evaluations	Independent of the leading model developers	Supports reproducible comparison and AutoML analysis on small-to-medium tables
Shmuel et al. [149]	A broad comparison of machine and deep learning across diverse datasets	Independent of the leading model developers	Reinforces that tuned trees are hard to beat off the small-IID regime
OpenML-CC18 and public OpenML [150]	Curated classification suites, mostly small-to-medium IID	Community, independent of the model developers	Independent but small-IID; carries a contamination risk for real-data-pretrained models
TabReD [151]	Industry-grade tables with temporal splits and documented pitfalls	Independent of the leading model developers	Tests temporal shift; a small collection

8.7. The Independent Evidence in Numbers

Before drawing the axes together, it is useful to state the evidence base in aggregate because the strength of the verdict differs sharply between axes and the difference is easy to lose in the prose. Table 11 records, for each axis, how many independent studies were retained, the largest evaluation behind the finding, and the direction in which that evidence points. The asymmetry is the substantive result. Generalisation and benchmark provenance rest on several studies and on evaluations spanning hundreds of datasets, so the finding that the paradigm cedes outside the small-to-medium IID regime is the best supported statement in this section. Calibration rests on one dedicated benchmark of 112 datasets and one scoring-rule study, which is enough to establish the trade-off and not enough to characterise it. Fairness rests on four audits, none of them large. Generation quality rests on a single comparative study, and the corresponding verdict in Figure 8 should be read as the weakest in the set. Any claim in this review therefore inherits the weight of the evidence behind its axis, and the table is the place to check that weight.

Table 11. The independent evidence behind each axis of this section, in aggregate. Study counts are distinct works retained for the axis; the evaluation scale is the largest cited for that axis. IID = independent and identically distributed; GBDT = gradient-boosted decision tree.

Axis	Studies	Largest Evaluation Cited	Direction of the Evidence
Generalisation beyond curated benchmarks	5	142 curated datasets across 11 models; a separate comparison over 300+ datasets	Tree-based and deep models lead outside the small-to-medium IID regime
Calibration and uncertainty	2	112 datasets of the TALENT suite, 4 foundation models	Highest accuracy with lower conditional coverage than GBDTs
Robustness under feature shift	4	Feature-shift and temporal-shift benchmarks in open environments	Accuracy degrades as the feature space moves; masking helps only under leakage control
Fairness	4	Standard algorithmic-fairness benchmarks; no large-scale audit	Group disparities survive pretraining; an explicit intervention is required
Interpretability	3	Method papers, no cross-model benchmark	Retrofitted post hoc or bought with an architectural constraint
Generation quality	1	One multi-table fidelity comparison	Trails diffusion generators; the weakest evidence base here
Benchmark provenance	10	7 independently assembled suites (Table 10)	Co-developed margins are weaker claims than independently reproduced ones

8.8. Trustworthiness: An Integrated Reading

The preceding subsections examined calibration, feature-shift robustness and fairness one axis at a time, and interpretability was treated with the capabilities in Section 5.4. Read together, these four form a single account of how far the paradigm can be trusted in a decision-facing deployment, and the account is more coherent than the axis-by-axis treatment suggests because the same structural feature of the paradigm underlies all four findings.

That feature is the amortised forward pass itself. A classical pipeline exposes intermediate objects that a practitioner can inspect and correct: a validation curve, a fitted model whose parameters can be examined, a resampling scheme that can be made shift-aware, and a calibration step that can be fitted separately. The in-context pass replaces all of these with one mapping from a labelled context and a query to a posterior predictive, learned from a synthetic prior. Whatever the prior did not encode is therefore absent from the output, and none of the four properties is recoverable from the pass by inspection. Calibrated conditional coverage is not encoded, so it must be supplied by an explicit conformal step [133]. Invariance to a changing feature space is not encoded because the pass treats the query as drawn like the context, so accuracy degrades when that assumption fails [135,136]. Group fairness is not encoded because a causal prior over synthetic tables carries no notion of a protected attribute, so disparities survive pretraining [137,138]. And the computation is distributed across attention over the context instead of being carried by an inspectable structure, so interpretability is retrofitted through post hoc attribution or through architectures constrained to an additive form, each at a cost in accuracy or generality [58,70,71].

Two practical conclusions follow, and they are stronger than any single axis supports on its own. First, the four properties are not independent problems to be solved separately but one problem seen from four directions: they are the properties a Bayesian posterior predictive was expected to carry and does not, and progress on them will come from what the prior encodes and how the readout is built, not from further scaling of the same recipe. The mechanistic audit of Section 5.2 supports this reading since the fragilities it induces are specific to the readout architecture [63]. Second, for the practitioner the four

collapse into one operational rule: a tabular foundation model should be treated as a strong accuracy component and not as a trustworthy decision system, so any deployment that carries consequences for people requires the explicit steps listed in Table 8, whatever the leaderboard position of the base model. Trustworthy use of the paradigm is, on the present evidence, a property of the surrounding pipeline and not of the model.

8.9. Open Problems and a Research Agenda

The open ends of the preceding subsections form a research agenda, set against the capability axes of Section 5. Seven problems stand out. (i) Generalisation: characterise where the accuracy of the paradigm crosses over to tree-based and deep models along the axes of distribution shift, scale and dimensionality, and close the gap on the non-IID regimes BeyondArena isolates. (ii) Calibration: recover the conditional coverage the models lose relative to trees, so that the posterior predictive they emit is usable for decisions as well as for ranking. (iii) Feature-shift robustness: make the forward pass resilient to features that shift or vanish at test time, under leakage-controlled evaluation. (iv) Fairness: build the fairness intervention into the pretraining or the in-context readout instead of leaving it to a downstream preprocessing step. (v) Generation fidelity: close the gap to diffusion generators on multivariate structure and on privacy before the language-model route is used for synthetic data in earnest. (vi) Context-length economics: quantify the accuracy, calibration and cost of the context-scaling techniques of Section 5.1 across the operating envelopes the developers report, which are stated ranges and not independently measured limits. (vii) Independent evaluation: report the headline capabilities of Section 5 on suites assembled outside the model-developing laboratories, so that the claims this review has tested can be confirmed on ground the developers did not choose. Each item is a capability recorded in Section 5 that the current evidence leaves unproven, and together they mark the distance between a working paradigm and a foundational one.

Table 12 restates the agenda as a problem-to-fix reference: each limit is paired with the condition that triggers it, the study that measures it, and its current mitigation, which is either an explicit external step or, more often, still open. Read as a whole, it is the review's central claim in operational form, the set of conditions under which the reported capabilities have not yet been shown to hold, each with the step that would close the gap.

Table 12. The failure modes synthesised in Section 8, the condition under which each appears, the study that measures it, and the current mitigation. The table complements the claim-to-evidence Table 9 with a problem-to-fix reading.

Failure Mode	When It Appears	Evidence	Mitigation Status
Generalisation loss	Non-IID, large or high-dimensional data	[121,152]	Open; tree-based and deep models remain ahead
Low conditional coverage	A decision needs a coverage guarantee	[133]	Partial; an external conformal step
Feature-shift accuracy loss	Features shift or vanish at test time	[135,136]	Open; train-time masking helps only under leakage control
Fairness disparity	Sensitive attributes are present	[137,138]	Partial; an explicit preprocessing step [139]
Generation fidelity gap	Multi-table structure, numeric fidelity, privacy	[141]	Open; diffusion generators remain ahead
Context-length degradation	Very long context windows	[62]	Partial; chunking or retrieval, at the cost of a lossy summary
Benchmark contamination	Real-data pretraining overlaps the evaluation	[121]	Open; read against independently assembled suites

9. Conclusions

The current generation of tabular foundation models has established amortised in-context prediction as a working paradigm. Across the 2024–2026 release train surveyed here, a pretrained transformer fits a small-to-medium table in a single forward pass and reaches the best reported results on curated benchmarks, and the mechanism has been carried beyond classification and regression to survival, causal, time-series, streaming, graph and anomaly tasks, though Figure 4 shows that this breadth is so far mostly an extension of the synthetic-prior line. The independent evidence assembled in Section 8 has not yet matched the benchmark claims, and the weight of that evidence differs by axis. It is strongest on generalisation, where five studies, the largest spanning 142 curated datasets across eleven models and a separate comparison covering more than three hundred datasets, place tree-based and deep models ahead outside the small-to-medium IID regime. It is sufficient but narrow on calibration, where a dedicated benchmark over the 112 datasets of the TALENT suite reports the highest accuracy together with lower conditional coverage than gradient-boosted trees. It is thinner on robustness under feature shift and on fairness, each carried by four studies and none of the fairness audits large. It rests on a single comparative study for generation quality, which is the weakest verdict recorded here. Table 11 states these quantities together, and each conclusion in this review should be read with the weight of its axis attached.

Two qualifications follow from how that evidence was assembled and should travel with every statement above. The synthesis reflects the corpus described in Section 2 and not the field in its entirety, and the corpus favours visible model lines and reported findings, so the picture it yields is, if anything, more favourable to the paradigm than the underlying literature may warrant. Two in five of the works cited are preprints or technical reports, concentrated in the most recent material, so a share of the evidence on both sides of the comparison has not been through external review. The conclusions here are accordingly a reading of the evidence available in mid-2026 and not a settled verdict on the paradigm.

Within those bounds, the practical consequence is a redistribution of effort rather than its removal. For practitioners the capability map is actionable on small IID tables, where the paradigm is strongest, and Section 7.1 sets out what changes around it: algorithm selection and hyperparameter search are absorbed, so problem formulation, data quality and context construction carry more of the weight, and validation acquires an explicit reliability step because Section 8.8 shows that calibration, shift robustness, fairness and interpretability are properties of the surrounding pipeline and not of the forward pass. For model developers the claim-versus-evidence spine marks the distance that separates a working paradigm from a foundational one: generalisation beyond curated benchmarks, calibrated uncertainty, robustness under feature shift, fairness, and evaluation on suites assembled outside the model-developing laboratories. The first four are, on the reading of Section 8.8, one problem seen from four directions since each names something the synthetic prior does not encode and the readout cannot recover; progress on them is likely to come from the design of priors and readouts and not from further scaling of the present recipe. The fifth is a matter of evaluation practice, and it is the one the community can settle soonest, by reporting the generation on suites it did not help to define.

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review and editing, V.S.T., A.S.B., V.A.N. and T.C.; visualisation, S.O.K.; supervision, V.S.T. and V.A.N.; project administration, V.S.T. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The screening corpus (961 records with per-record inclusion decisions), the Scopus query, and the model summary underlying Tables 3 and 7 (models, versions, licences, weight and code links, accessed dates) are deposited at Zenodo: https://doi.org/10.5281/zenodo.21416877.

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Appendix A. Search and Screening Strategy

The corpus was assembled from a broad identification query on Scopus, run over 2023–2026 and restricted to the computer-science and engineering subject areas and to the article, review and conference-paper document types. The query paired the term *tabular* with the vocabulary of the paradigm and was deliberately wide at the identification stage:

> (TITLE-ABS-KEY (tabular) AND TITLE-ABS-KEY (“foundation model*” OR “in-context learning” OR “prior-data fitted” OR “TabPFN” OR “tabular transformer*” OR “large language model*” OR “pre-trained model*” OR “pretrained model*”)) AND PUBYEAR > 2022 AND PUBYEAR < 2027 AND (LIMIT-TO (DOCTYPE , “ar”) OR LIMIT-TO (DOCTYPE , “re”) OR LIMIT-TO (DOCTYPE , “cp”)) AND (LIMIT-TO (SUBJAREA , “COMP”) OR LIMIT-TO (SUBJAREA , “ENGI”))

Records were completed and cross-checked against two further public sources over the same period. The arXiv corpus was searched at the abstract level to locate preprints and to verify identifiers and versions:

> (abs:“tabular” AND (abs:“foundation model” OR abs:“in-context learning” OR abs:“prior-data fitted” OR abs:“TabPFN” OR abs:“tabular transformer” OR abs:“tabular deep learning”)) AND (cat:cs.LG OR cat:stat.ML OR cat:cs.AI) AND submittedDate:[20230101 TO 20261231]

The DBLP computer-science bibliography was used to confirm author lists, venues and canonical records, by per-title lookup at https://dblp.org/search?q= (accessed on 21 July 2026) and by the following keyword searches, complemented by browsing the 2023–2026 records of the main venues (NeurIPS, ICML, ICLR):

> `tabular foundation model`; `TabPFN`; `in-context learning tabular`; `prior-data fitted network`; `tabular deep learning`

The returned records were screened in two passes against the scope of Section 2. A title-and-abstract pass removed table-question-answering, natural-language-interface and table-extraction work, and a full-text pass on the retained records assigned each to a section of the review or excluded it as off-scope. As stated in Section 2, this is a narrative review and no completeness claim of the PRISMA kind is made; the counts are reported for transparency. The screening decisions, the query, the model summary, and a machine-readable index of the bibliography giving an arXiv and a DBLP link for each cited work are deposited at the repository named in the Data Availability Statement; every cited work carries an arXiv identifier or a DOI, verified against the arXiv and DBLP records.

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